ELEC 2FH4 – Topics + HW Reference

In Class Topics Cross Referenced w/ Textbook

Vector Algebra

- Scalars and Vectors (1.3 - p. 4)
- Unit Vector (1.4 - p. 5)
- Vector Addition and Subtraction (1.5 - p. 6)
- Position and Distance Vectors (1.6 - p. 7)
- Vector Multiplication (1.7 - p. 11)
- Components of a Vector (1.8 – p. 16)

Coordinate Systems

- Vector triple product (1.7 – p. 11)
- Components of a vector (1.8 – p. 16)
- Circular cylindrical coordinates (2.3 – p. 32)
- Spherical Coordinates (2.4 – p. 35)
- Constant-Coordinate Surfaces (2.5 – p. 44)

Vector Calculus

- Differential Length, Area, and Volume (3.2 – p. 59)
- Line, Surface, and Volume Integrals (3.3 – p. 66)
- Fields, Scalar and Vector
- Del Operator (3.4 – p. 69)
- Gradient of a Scalar (3.5 – p. 71)
- Divergence of a Vector (3.6 – p. 75)
- Divergence Theorem (3.6 – p. 78)
- Curl of a Vector and Stokes's Theorem (3.7 – p. 82)
- Classification of Vector Fields (3.9 – p. 92)
- Laplacian (3.8 – p. 90)
- Divergence and Curl with scaling factors
- Chapter Review

Electrostatics

- Coulomb's Law and Field Intensity (4.2 – p. 112)

- Electric Fields Due to Continuous Charge Distribution (4.3 – p. 119)
- Electric Flux Density (4.4 – p. 130)
- Gauss' Law—Maxwell's Equation (4.5 – p. 132)
- Applications of Gauss' Law (4.6 – p. 134)
- Electric Potential (4.7 – p. 141)
- Relationship between E and V—Maxwell's Equation (4.8 – p. 147)
- An Electric Dipole and Flux Lines (4.9 – p. 150)
- Energy Density in Electrostatic Fields (4.10 – p. 154)
- Convection and Conduction Currents (5.3 – p. 178)
- Conductors (5.4 – p. 181)
- Conductors (5.4 – p. 181)
- Polarization in Dielectrics (5.5 – p. 187)
- Dielectric Constant and Strength (5.6 – p. 190)
- Continuity Equation and Relaxation Time (5.8 – p. 196)
- Boundary Conditions (5.9 – p. 198)
- Poisson's and Laplace's Equations (6.2 – p. 225)
- Solving Poisson's or Laplace's Equation (6.4 – p. 228)
- Resistance and Capacitance (6.5 – p. 249)
- Method of Images (6.6 – p. 266)

Magnetostatics

- Biot–Savart's Law (7.2 – p. 298)
- Ampère's Circuit Law—Maxwell's Equation (7.3 – p. 309)
- Applications of Ampère's Law (7.4 – p. 309)
- Magnetic Flux Density—Maxwell's Equation (7.5 – p. 317)
- Maxwell's Equations for Static Fields (7.6 – p. 319)
- Magnetic Scalar and Vector Potentials (7.7 – p. 320
- Forces due to Magnetic Fields (8.2 – p. 349)
- Magnetic Torque and Moment (8.3 – p. 361) <----- we're here at the moment

Textbook HW

Chapter 1

- Practice Exercises: 1.2, 1.4, 1.5
- End-of-Chapter Problems: 1.3, 1.4, 1.6, 1.8, 1.11, 1.13, 1.19

Chapter 2

- End-of-Chapter Problems: 2.1 – 2.6, 2.11, 2.14, 2.15, 2.17, 2.18, 2.29

Chapter 3

- End-of-Chapter Problems: 3.1 – 3.6, 3.11, 3.16, 3.17, 3.18, 3.20, 3.27 – 3.29, 3.31 – 3.36, 3.40 – 3.43, 3.45 – 3.48, 3.51, 3.52, 3.54, 3.55 – 3.60
- Problem 3.36 is especially important

Chapter 4

- End-of-Chapter Problems: 4.1, 4.2, 4.3, 4.5 – 4.7
- Sections 4-5 to 4-8

Chapter 5

- End-of-Chapter Problems: 5.1–5.5, 5.8, 5.14–5.19, 5.30, 5.31, 5.38–5.42

Chapter 6

- End-of-Chapter Problems: 6.1, 6.2, 6.4, 6.7, 6.11, 6.13–6.16, 6.29–6.31, 6.66

Chapter 7

- End-of-Chapter Problems: 7.1–7.4, 7.6, 7.7, 7.10, 7.16, 7.17, 7.34, 7.37, 7.40, 7.41

Chapter 8

- End-of-Chapter Problems: 8.1–8.9, 8.18–8.21, 8.24–8.26, 8.30–8.34, 8.42–8.46, 8.50, 8.51

Chapter 9

- oopsies, we don’t cover this

Chapter 10

- oopsies, we don’t cover this

Textbook Chapter Summaries

Chapter 1

SUMMARY

1. A field is a function that specifies a quantity in space. For example, $\mathbf{A}(x, y, z)$ is a vector field, whereas $V(x, y, z)$ is a scalar field.
2. A vector $\mathbf{A}$ is uniquely specified by its magnitude and a unit vector along it, that is, $\mathbf{A} = A\mathbf{a}_A$.
3. Multiplying two vectors $\mathbf{A}$ and $\mathbf{B}$ results in either a scalar $\mathbf{A} \cdot \mathbf{B} = AB \cos \theta_{AB}$ or a vector $\mathbf{A} \times \mathbf{B} = AB \sin \theta_{AB}\, \mathbf{a}_n$. Multiplying three vectors $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ yields a scalar $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$ or a vector $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$.
4. The scalar projection (or component) of vector $\mathbf{A}$ onto $\mathbf{B}$ is $A_B = \mathbf{A} \cdot \mathbf{a}_B$, whereas vector projection of $\mathbf{A}$ onto $\mathbf{B}$ is $\mathbf{A}_B = A_B\mathbf{a}_B$.
5. The MATLAB commands dot(A,B) and cross(A,B) are used for dot and cross products, respectively.

Chapter 2

SUMMARY

1. The three common coordinate systems we shall use throughout the text are the Cartesian (or rectangular), the circular cylindrical, and the spherical.
2. A point P is represented as $P(x, y, z)$, $P(\rho, \phi, z)$, and $P(r, \theta, \phi)$ in the Cartesian, cylindrical, and spherical systems, respectively. A vector field $\mathbf{A}$ is represented as (A_x, A_y, A_z) or $A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z$ in the Cartesian system, as (A_ρ, A_ϕ, A_z) or $A_\rho\mathbf{a}_\rho + A_\phi\mathbf{a}_\phi + A_z\mathbf{a}_z$ in the cylindrical system, and as (A_r, A_θ, A_ϕ) or $A_r\mathbf{a}_r + A_\theta\mathbf{a}_\theta + A_\phi\mathbf{a}_\phi$ in the spherical system. It is preferable that mathematical operations (addition, subtraction, product, etc.) be performed in the same coordinate system. Thus, point and vector transformations should be performed whenever necessary. A summary of point and vector transformations is given in Table 2.1.
3. Fixing one space variable defines a surface; fixing two defines a line; fixing three defines a point.
4. A unit normal vector to surface n = constant is $\pm\mathbf{a}_n$.

TABLE 2.1 Relationships between Rectangular, Cylindrical, and Spherical Coordinates

Rectangular to Cylindrical	Cylindrical to Rectangular
Variable change $\begin{cases} x = \rho\cos\phi \\ y = \rho\sin\phi \\ z = z \end{cases}$	Variable change $\begin{cases} \rho = \sqrt{x^2+y^2} \\ \phi = \tan^{-1}\left(\dfrac{y}{x}\right) \begin{cases} \sin\phi = \dfrac{y}{\sqrt{x^2+y^2}} \\ \cos\phi = \dfrac{x}{\sqrt{x^2+y^2}} \end{cases} \\ z = z \end{cases}$
Component change $\begin{cases} A_\rho = A_x\cos\phi + A_y\sin\phi \\ A_\phi = -A_x\sin\phi + A_y\cos\phi \\ A_z = A_z \end{cases}$	Component change $\begin{cases} A_x = A_\rho\dfrac{x}{\sqrt{x^2+y^2}} - A_\phi\dfrac{y}{\sqrt{x^2+y^2}} \\ A_y = A_\rho\dfrac{y}{\sqrt{x^2+y^2}} + A_\phi\dfrac{x}{\sqrt{x^2+y^2}} \\ A_z = A_z \end{cases}$
Rectangular to Spherical	**Spherical to Rectangular**
Variable change $\begin{cases} x = r\sin\theta\cos\phi \\ y = r\sin\theta\sin\phi \\ z = r\cos\theta \end{cases}$	Variable change $\begin{cases} r = \sqrt{x^2+y^2+z^2} \\ \theta = \cos^{-1}\dfrac{z}{\sqrt{x^2+y^2+z^2}} \begin{cases} \cos\theta = \dfrac{z}{\sqrt{x^2+y^2+z^2}} \\ \sin\theta = \dfrac{\sqrt{x^2+y^2}}{\sqrt{x^2+y^2+z^2}} \end{cases} \\ \phi = \tan^{-1}\left(\dfrac{y}{x}\right) \begin{cases} \cos\phi = \dfrac{x}{\sqrt{x^2+y^2}} \\ \sin\phi = \dfrac{y}{\sqrt{x^2+y^2}} \end{cases} \end{cases}$
Component change $\begin{cases} A_r = A_x\sin\theta\cos\phi + A_y\sin\theta\sin\phi + A_z\cos\theta \\ A_\theta = A_x\cos\theta\cos\phi + A_y\cos\theta\sin\phi - A_z\sin\theta \\ A_\phi = -A_x\sin\phi + A_y\cos\phi \end{cases}$	Component change $\begin{cases} A_x = \dfrac{A_r x}{\sqrt{x^2+y^2+z^2}} + \dfrac{A_\theta xz}{\sqrt{(x^2+y^2)(x^2+y^2+z^2)}} - \dfrac{A_\phi y}{\sqrt{x^2+y^2}} \\ A_y = \dfrac{A_r y}{\sqrt{x^2+y^2+z^2}} + \dfrac{A_\theta yz}{\sqrt{(x^2+y^2)(x^2+y^2+z^2)}} + \dfrac{A_\phi x}{\sqrt{x^2+y^2}} \\ A_z = \dfrac{A_r z}{\sqrt{x^2+y^2+z^2}} - \dfrac{A_\theta\sqrt{x^2+y^2}}{\sqrt{x^2+y^2+z^2}} \end{cases}$

Chapter 3

SUMMARY

1. The differential displacements in the Cartesian, cylindrical, and spherical systems are, respectively,

$$d\mathbf{l} = dx\,\mathbf{a}_x + dy\,\mathbf{a}_y + dz\,\mathbf{a}_z$$
$$d\mathbf{l} = d\rho\,\mathbf{a}_\rho + \rho\,d\phi\,\mathbf{a}_\phi + dz\,\mathbf{a}_z$$
$$d\mathbf{l} = dr\,\mathbf{a}_r + r\,d\theta\,\mathbf{a}_\theta + r\sin\theta\,d\phi\,\mathbf{a}_\phi$$

Note that $d\mathbf{l}$ is always taken to be in the positive direction; the direction of the displacement is taken care of by the limits of integration.

2. The differential normal areas in the three systems are, respectively,

$$\begin{aligned} d\mathbf{S} &= dy\,dz\,\mathbf{a}_x \\ &\quad dx\,dz\,\mathbf{a}_y \\ &\quad dx\,dy\,\mathbf{a}_z \end{aligned}$$

$$\begin{aligned} d\mathbf{S} &= \rho\,d\phi\,dz\,\mathbf{a}_\rho \\ &\quad d\rho\,dz\,\mathbf{a}_\phi \\ &\quad \rho\,d\rho\,d\phi\,\mathbf{a}_z \end{aligned}$$

$$\begin{aligned} d\mathbf{S} &= r^2\sin\theta\,d\theta\,d\phi\,\mathbf{a}_r \\ &\quad r\sin\theta\,dr\,d\phi\,\mathbf{a}_\theta \\ &\quad r\,dr\,d\theta\,\mathbf{a}_\phi \end{aligned}$$

Note that $d\mathbf{S}$ can be in the positive or negative direction depending on the surface under consideration.

3. The differential volumes in the three systems are

$$dv = dx\,dy\,dz$$
$$dv = \rho\,d\rho\,d\phi\,dz$$
$$dv = r^2\sin\theta\,dr\,d\theta\,d\phi$$

4. The line integral of vector $\mathbf{A}$ along a path L is given by $\int_L \mathbf{A}\cdot d\mathbf{l}$. If the path is closed, the line integral becomes the circulation of $\mathbf{A}$ around L, that is, $\oint_L \mathbf{A}\cdot d\mathbf{l}$.
5. The flux or surface integral of a vector $\mathbf{A}$ across a surface S is defined as $\int_S \mathbf{A}\cdot d\mathbf{S}$. When the surface S is closed, the surface integral becomes the net outward flux of $\mathbf{A}$ across S, that is, $\oint_S \mathbf{A}\cdot d\mathbf{S}$.
6. The volume integral of a scalar ρ_v over a volume v is defined as $\int_v \rho_v\,dv$.
7. Vector differentiation is performed by using the vector differential operator ∇. The gradient of a scalar field V is denoted by ∇V, the divergence of a vector field $\mathbf{A}$ by $\nabla\cdot\mathbf{A}$, the curl of $\mathbf{A}$ by $\nabla\times\mathbf{A}$, and the Laplacian of V by $\nabla^2 V$. All of these are point functions since differentiation is always at a point.
8. The divergence theorem, $\oint_S \mathbf{A}\cdot d\mathbf{S} = \int_v \nabla\cdot\mathbf{A}\,dv$, relates a surface integral over a closed surface to a volume integral.
9. Stokes's theorem, $\oint_L \mathbf{A}\cdot d\mathbf{l} = \int_S (\nabla\times\mathbf{A})\cdot d\mathbf{S}$, relates a line integral over a closed path to a surface integral.
10. If Laplace's equation, $\nabla^2 V = 0$, is satisfied by a scalar field V in a given region, V is said to be harmonic in that region.
11. A vector field is solenoidal if $\nabla\cdot\mathbf{A} = 0$; it is irrotational or conservative if $\nabla\times\mathbf{A} = \mathbf{0}$.
12. A summary of the vector calculus operations in the three coordinate systems is provided on the inside back cover of the text.
13. The vector identities $\nabla\cdot\nabla\times\mathbf{A} = 0$ and $\nabla\times\nabla V = \mathbf{0}$ are very useful in EM. Other vector identities are in Appendix A.10.

Chapter 4

SUMMARY

1. The two fundamental laws for electrostatic fields (Coulomb's and Gauss's) are presented in this chapter. Coulomb's law of force states that

$$\mathbf{F} = \frac{Q_1 Q_2}{4\pi\varepsilon_o R^2}\,\mathbf{a}_R$$

2. Based on Coulomb's law, we define the electric field intensity **E** as the force per unit charge; that is,

$$\mathbf{E} = \frac{Q}{4\pi\varepsilon_o R^2}\,\mathbf{a}_R = \frac{Q\,\mathbf{R}}{4\pi\varepsilon_o R^3} \quad \text{(point charge only)}$$

3. For a continuous charge distribution, the total charge is given by

$$Q = \int_L \rho_L\, dl \qquad \text{for line charge}$$

$$Q = \int_S \rho_S\, dS \qquad \text{for surface charge}$$

$$Q = \int_v \rho_v\, dv \qquad \text{for volume charge}$$

The **E** field due to a continuous charge distribution is obtained from the formula for point charge by replacing Q with $dQ = \rho_L\, dl$, $dQ = \rho_S\, dS$ or $dQ = \rho_v\, dv$ and integrating over the line, surface, or volume, respectively.

4. For an infinite line charge,

$$\mathbf{E} = \frac{\rho_L}{2\pi\varepsilon_o \rho}\,\mathbf{a}_\rho$$

and for an infinite sheet of charge,

$$\mathbf{E} = \frac{\rho_S}{2\varepsilon_o}\mathbf{a}_n$$

5. The electric flux density $\mathbf{D}$ is related to the electric field intensity (in free space) as

$$\mathbf{D} = \varepsilon_o \mathbf{E}$$

The electric flux through a surface S is

$$\Psi = \int_S \mathbf{D} \cdot d\mathbf{S}$$

6. Gauss's law states that the net electric flux penetrating a closed surface is equal to the total charge enclosed, that is, $\Psi = Q_{enc}$. Hence,

$$\Psi = \oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{enc} = \int_v \rho_v \, dv$$

or

$$\rho_v = \nabla \cdot \mathbf{D} \quad \text{(first Maxwell equation to be derived)}$$

When charge distribution is symmetric, so that a Gaussian surface (where $\mathbf{D} = D_n \mathbf{a}_n$ is constant) can be found, Gauss's law is useful in determining $\mathbf{D}$; that is,

$$D_n \oint_S dS = Q_{enc} \quad \text{or} \quad D_n = \frac{Q_{enc}}{S}$$

7. The total work done, or the electric potential energy, to move a point charge Q from point A to B in an electric field $\mathbf{E}$ is

$$W = -Q \int_A^B \mathbf{E} \cdot d\mathbf{l}$$

8. The potential at $\mathbf{r}$ due to a point charge Q at $\mathbf{r}'$ is

$$V(\mathbf{r}) = \frac{Q}{4\pi\varepsilon_o |\mathbf{r} - \mathbf{r}'|} + C$$

where C is evaluated at a given reference potential point; for example, $C = 0$ if $V(\mathbf{r} \to \infty) = 0$. To determine the potential due to a continuous charge distribution, we replace Q in the formula for point charge by $dQ = \rho_L \, dl$, $dQ = \rho_S \, dS$, or $dQ = \rho_v \, dv$ and integrate over the line, surface, or volume, respectively.

9. If the charge distribution is not known, but the field intensity **E** is given, we find the potential by using

$$V = -\int_L \mathbf{E} \cdot d\mathbf{l} + C$$

10. The potential difference V_{AB}, the potential at B with reference to A, is

$$V_{AB} = -\int_A^B \mathbf{E} \cdot d\mathbf{l} = \frac{W}{Q} = V_B - V_A$$

11. Since an electrostatic field is conservative (the net work done along a closed path in a static **E** field is zero),

$$\oint_L \mathbf{E} \cdot d\mathbf{l} = 0$$

or

$$\nabla \times \mathbf{E} = \mathbf{0} \quad \text{(second Maxwell equation to be derived)}$$

12. Given the potential field, the corresponding electric field is found by using

$$\mathbf{E} = -\nabla V$$

13. For an electric dipole centered at $\mathbf{r}'$ with dipole moment $\mathbf{p}$, the potential at $\mathbf{r}$ is given by

$$V(\mathbf{r}) = \frac{\mathbf{p} \cdot (\mathbf{r} - \mathbf{r}')}{4\pi\varepsilon_o |\mathbf{r} - \mathbf{r}'|^3}$$

14. The flux density **D** is tangential to the electric flux lines at every point. An equipotential surface (or line) is one on which V = constant. At every point, the equipotential line is orthogonal to the electric flux line.
15. The electrostatic energy due to n point charges is

$$W_E = \frac{1}{2} \sum_{k=1}^{n} Q_k V_k$$

For a continuous volume charge distribution,

$$W_E = \frac{1}{2} \int_v \mathbf{D} \cdot \mathbf{E}\, dv = \frac{1}{2} \int_v \varepsilon_o |\mathbf{E}|^2\, dv$$

16. Electrostatic discharge (ESD) refers to the sudden transfer of static charge between objects at different electrostatic potentials. Since all semiconductor devices are regarded as ESD sensitive, a good understandng of ESD is required in industry.

Chapter 5

SUMMARY

1. Materials can be classified roughly as conductors ($\sigma \gg 1, \varepsilon_r = 1$) and dielectrics ($\sigma \ll 1, \varepsilon_r \geq 1$) in terms of their electrical properties σ and ε_r, where σ is the conductivity and ε_r is the dielectric constant or relative permittivity.
2. Electric current is the flux of electric current density through a surface; that is,

$$I = \int \mathbf{J} \cdot d\mathbf{S}$$

3. The resistance of a conductor of uniform cross section is

$$R = \frac{\ell}{\sigma S}$$

4. The macroscopic effect of polarization on a given volume of a dielectric material is to "paint" its surface with a bound charge $Q_b = \oint_S \rho_{ps}\, dS$ and leave within it an accumulation of bound charge $Q_b = \int_v \rho_{pv}\, dv$, where $\rho_{ps} = \mathbf{P} \cdot \mathbf{a}_n$ and $\rho_{pv} = -\nabla \cdot \mathbf{P}$.
5. In a dielectric medium, the **D** and **E** fields are related as $\mathbf{D} = \varepsilon\mathbf{E}$, where $\varepsilon = \varepsilon_o\varepsilon_r$ is the permittivity of the medium while **E** and **P** are related as $\mathbf{P} = \chi_e\varepsilon_o\mathbf{E}$.
6. The electric susceptibility $\chi_e (= \varepsilon_r - 1)$ of a dielectric measures the sensitivity of the material to an electric field.
7. A dielectric material is linear if $\mathbf{D} = \varepsilon\mathbf{E}$ holds, that is, if ε is independent of **E**. It is homogeneous if ε is independent of position. It is isotropic if ε is a scalar.
8. The principle of charge conservation, the basis of Kirchhoff's current law, is stated in the continuity equation

$$\nabla \cdot \mathbf{J} + \frac{\partial \rho_v}{\partial t} = 0$$

9. The relaxation time, $T_r = \varepsilon/\sigma$, of a material is the time taken by a charge placed in its interior to decrease by a factor of e^{-1} or to $\simeq$37% of its original magnitude.
10. Boundary conditions must be satisfied by an electric field existing in two different media separated by an interface. For a dielectric–dielectric interface

$$E_{1t} = E_{2t}$$

$$D_{1n} - D_{2n} = \rho_S \quad \text{or} \quad D_{1n} = D_{2n} \quad \text{if} \quad \rho_S = 0$$

For a dielectric–conductor interface,

$$E_t = 0, \quad D_n = \varepsilon E_n = \rho_S$$

because $\mathbf{E} = \mathbf{0}$ inside the conductor.

11. Materials of high dielectric constant are of great interest for high-performance electronic devices.

Chapter 6

SUMMARY

1. Boundary-value problems are those in which the potentials or their derivatives at the boundaries of a region are specified and we are to determine the potential field within the region. They are solved by using Poisson's equation if $\rho_v \neq 0$ or Laplace's equation if $\rho_v = 0$.
2. In a nonhomogeneous region, Poisson's equation is

$$\nabla \cdot \varepsilon \, \nabla V = -\rho_v$$

   For a homogeneous region, ε is independent of space variables. Poisson's equation becomes

$$\nabla^2 V = -\frac{\rho_v}{\varepsilon}$$

   In a charge-free region $(\rho_v = 0)$, Poisson's equation becomes Laplace's equation; that is,

$$\nabla^2 V = 0$$

3. We solve the differential equation resulting from Poisson's or Laplace's equation by integrating twice if V depends on one variable or by the method of separation of variables if V is a function of more than one variable. We then apply the prescribed boundary conditions to obtain a unique solution.

4. The uniqueness theorem states that if V satisfies Poisson's or Laplace's equation and the prescribed boundary condition, V is the only possible solution for that problem. This enables us to find the solution to a given problem via any expedient means because we are assured of one, and only one, solution.
5. The problem of finding the resistance R of an object or the capacitance C of a capacitor may be treated as a boundary-value problem. To determine R, we assume a potential difference V_o between the ends of the object, solve Laplace's equation, find $I = \int_S \sigma \mathbf{E} \cdot d\mathbf{S}$, and obtain $R = V_o/I$. Similarly, to determine C, we assume a potential difference of V_o between the plates of the capacitor, solve Laplace's equation, find $Q = \int_S \varepsilon \mathbf{E} \cdot d\mathbf{S}$, and obtain $C = Q/V_o$.
6. A boundary-value problem involving an infinite conducting plane or wedge may be solved by using the method of images. This basically entails replacing the charge configuration by itself, its image, and an equipotential surface in place of the conducting plane. Thus the original problem is replaced by "an image problem," which is solved by using techniques covered in Chapters 4 and 5.
7. The computation of the capacitance of microstrip lines has become important because such lines are used in microwave devices. Three formulas for finding the capacitance of a circular microstrip line have been presented.

Recall that the Laplacian operator ∇^2 was derived in Section 3.8. Thus Laplace's equation in Cartesian, cylindrical, or spherical coordinates, respectively, is given by

$$\boxed{\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0} \tag{6.6}$$

$$\boxed{\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho\frac{\partial V}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} = 0} \tag{6.7}$$

$$\boxed{\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\,\frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 V}{\partial \phi^2} = 0} \tag{6.8}$$

depending on the coordinate variables used to express V, that is, $V(x, y, z)$, $V(\rho, \phi, z)$, or $V(r, \theta, \phi)$. Poisson's equation in those coordinate systems may be obtained by simply replacing zero on the right-hand side of eqs. (6.6), (6.7), and (6.8) with $-\rho_v/\varepsilon$.

Chapter 7

TABLE 7.1 Analogy between Electric and Magnetic Fields*

Term	Electric	Magnetic
Basic laws	$\mathbf{F} = \dfrac{Q_1Q_2}{4\pi\varepsilon R^2}\mathbf{a}_R$	$d\mathbf{B} = \dfrac{\mu_o I\, d\mathbf{l} \times \mathbf{a}_R}{4\pi R^2}$
	$\oint \mathbf{D} \cdot d\mathbf{S} = Q_{enc}$	$\oint \mathbf{H} \cdot d\mathbf{l} = I_{enc}$
Force law	$\mathbf{F} = Q\mathbf{E}$	$\mathbf{F} = Q\mathbf{u} \times \mathbf{B}$
Source element	dQ	$dQ\mathbf{u} = \mathrm{I}d\mathbf{l}$
Field intensity	$E = \dfrac{V}{\ell}$ (V/m)	$H = \dfrac{I}{\ell}$ (A/m)
Flux density	$D = \dfrac{\psi}{S}$ (C/m^2)	$B = \dfrac{\psi}{S}$ (Wb/m^2)
Relationship between fields	$\mathbf{D} = \varepsilon\mathbf{E}$	$\mathbf{B} = \mu\mathbf{H}$
Potentials	$\mathbf{E} = -\nabla V$	$\mathbf{H} = -\nabla V_m\ (\mathbf{J} = 0)$
	$V = \displaystyle\int_L \frac{\rho_L dl}{4\pi\varepsilon R}$	$\mathbf{A} = \displaystyle\int_L \frac{\mu\, I\, d\ell}{4\pi R}$
Flux	$\psi = \int \mathbf{D} \cdot d\mathbf{S}$	$\psi = \int_S \mathbf{B} \cdot d\mathbf{S}$
	$\psi = Q = CV$	$\psi = LI$
	$I = C\dfrac{dV}{dt}$	$V = L\dfrac{dI}{dt}$
Energy density	$w_E = \dfrac{1}{2}\mathbf{D} \cdot \mathbf{E}$	$w_m = \dfrac{1}{2}\mathbf{B} \cdot \mathbf{H}$
Poisson's equation	$\nabla^2 V = -\dfrac{\rho_v}{\varepsilon}$	$\nabla^2\mathbf{A} = -\mu\mathbf{J}$

SUMMARY

1. The basic laws (Biot–Savart's and Ampère's) that govern magnetostatic fields are discussed. Biot–Savart's law, which is similar to Coulomb's law, states that the magnetic field intensity $d\mathbf{H}$ at $\mathbf{r}$ due to current element $I\,d\mathbf{l}$ at $\mathbf{r}'$ is

$$d\mathbf{H} = \frac{I\,d\mathbf{l} \times \mathbf{R}}{4\pi R^3} \quad \text{(in A/m)}$$

where $\mathbf{R} = \mathbf{r} - \mathbf{r}'$ and $R = |\mathbf{R}|$. For surface or volume current distribution, we replace $I\,d\mathbf{l}$ with $\mathbf{K}\,dS$ or $\mathbf{J}\,dv$, respectively; that is,

$$I\,d\mathbf{l} \equiv \mathbf{K}\,dS \equiv \mathbf{J}\,dv$$

2. Ampère's circuit law, which is similar to Gauss's law, states that the circulation of $\mathbf{H}$ around a closed path is equal to the current enclosed by the path; that is,

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = I_{enc} = \int_S \mathbf{J} \cdot d\mathbf{S}$$

or

$$\nabla \times \mathbf{H} = \mathbf{J} \quad \text{(third Maxwell equation to be derived)}$$

When current distribution is symmetric so that an Amperian path (on which $\mathbf{H} = H_\phi \mathbf{a}_\phi$ is constant) can be found, Ampère's law is useful in determining $\mathbf{H}$; that is,

$$H_\phi \oint_L dl = I_{\text{enc}} \quad \text{or} \quad H_\phi = \frac{I_{\text{enc}}}{\ell}$$

3. The magnetic flux through a surface S is given by

$$\Psi = \int_S \mathbf{B} \cdot d\mathbf{S} \quad \text{(in Wb)}$$

where $\mathbf{B}$ is the magnetic flux density (in Wb/m^2). In free space,

$$\mathbf{B} = \mu_o \mathbf{H}$$

where $\mu_o = 4\pi \times 10^{-7}$ H/m = permeability of free space.

4. Since an isolated or free magnetic monopole does not exist, the net magnetic flux through a closed surface is zero:

$$\Psi = \oint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

or

$$\nabla \cdot \mathbf{B} = 0 \quad \text{(fourth Maxwell equation to be derived)}$$

5. At this point, all four Maxwell equations for static EM fields have been derived, namely:

$$\begin{aligned} \nabla \cdot \mathbf{D} &= \rho_v \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{E} &= \mathbf{0} \\ \nabla \times \mathbf{H} &= \mathbf{J} \end{aligned}$$

6. The magnetic scalar potential V_m is defined as

$$\mathbf{H} = -\nabla V_m, \quad \text{if } \mathbf{J} = \mathbf{0}$$

and the magnetic vector potential $\mathbf{A}$ as

$$\mathbf{B} = \nabla \times \mathbf{A}$$

where $\nabla \cdot \mathbf{A} = 0$. With the definition of $\mathbf{A}$, the magnetic flux through a surface S can be found from

$$\Psi = \oint_L \mathbf{A} \cdot d\mathbf{l}$$

where L is the closed path defining surface S (see Figure 3.21). Rather than using Biot–Savart's law, the magnetic field due to a current distribution may be found by using $\mathbf{A}$, a powerful approach that is particularly useful in antenna theory. For a current element $I\,d\mathbf{l}$ at $\mathbf{r}'$, the magnetic vector potential at $\mathbf{r}$ is

$$\mathbf{A} = \int \frac{\mu_o I\, d\mathbf{l}}{4\pi R}, \quad R = |\mathbf{r} - \mathbf{r}'|$$

7. Elements of similarity between electric and magnetic fields exist. Some of these are listed in Table 7.1. Corresponding to Poisson's equation $\nabla^2 V = -\rho_v/\varepsilon$, for example, is

$$\nabla^2 \mathbf{A} = -\mu_o \mathbf{J}$$

8. Lightning may be regarded as a transient, high-current electric discharge. A common way to protect people, buildings, and other structures from lightning is to use lightning rods.

REVIEW QUESTIONS

7.1 One of the following is not a source of magnetostatic fields:

(a) A dc current in a wire
(b) A permanent magnet
(c) An accelerated charge
(d) An electric field linearly changing with time
(e) A charged disk rotating at uniform speed

7.2 Identify the configuration in Figure 7.24 that is not a correct representation of I and $\mathbf{H}$.

7.3 Consider points A, B, C, D, and E on a circle of radius 2 as shown in Figure 7.25. The items in the right-hand list are the values of $\mathbf{a}_\phi$ at different points on the circle. Match these items with the points in the list on the left.

(a) A (i) $\mathbf{a}_x$
(b) B (ii) $-\mathbf{a}_x$
(c) C (iii) $\mathbf{a}_y$
(d) D (iv) $-\mathbf{a}_y$

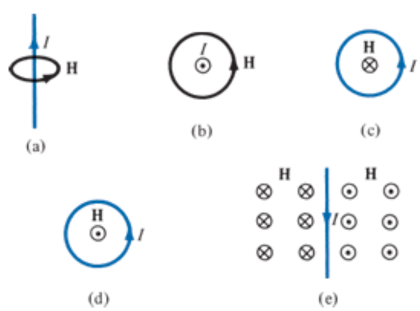


FIGURE 7.24 For Review Question 7.2.

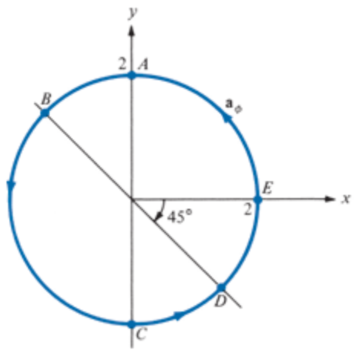


FIGURE 7.25 For Review Question 7.3.

(e) E (v) $\dfrac{\mathbf{a}_x + \mathbf{a}_y}{\sqrt{2}}$

(vi) $\dfrac{-\mathbf{a}_x - \mathbf{a}_y}{\sqrt{2}}$

(vii) $\dfrac{-\mathbf{a}_x + \mathbf{a}_y}{\sqrt{2}}$

(viii) $\dfrac{\mathbf{a}_x - \mathbf{a}_y}{\sqrt{2}}$

7.4 The z-axis carries filamentary current of 10π A along $\mathbf{a}_z$. Which of these is incorrect?

(a) $\mathbf{H} = -\mathbf{a}_x$ A/m at $(0, 5, 0)$
(b) $\mathbf{H} = \mathbf{a}_\phi$ A/m at $(5, \pi/4, 0)$
(c) $\mathbf{H} = -0.8\mathbf{a}_x - 0.6\mathbf{a}_y$ at $(-3, 4, 0)$
(d) $\mathbf{H} = -\mathbf{a}_\phi$ at $(5, 3\pi/2, 0)$

7.5 Plane $y = 0$ carries a uniform current of $30\mathbf{a}_z$ mA/m. At $(1, 10, -2)$, the magnetic field intensity is

(a) $-15\mathbf{a}_x$ mA/m
(b) $15\mathbf{a}_x$ mA/m
(c) $477.5\mathbf{a}_y$ μA/m
(d) $18.85\mathbf{a}_y$ nA/m
(e) None of the above

7.6 For the currents and closed paths of Figure 7.26, calculate the value of $\oint_L \mathbf{H} \cdot d\mathbf{l}$.

7.7 Which of these statements is not characteristic of a static magnetic field?

(a) It is solenoidal.
(b) It is conservative.
(c) It has no sinks or sources.
(d) Magnetic flux lines are always closed.
(e) The total number of flux lines entering a given region is equal to the total number of flux lines leaving the region.

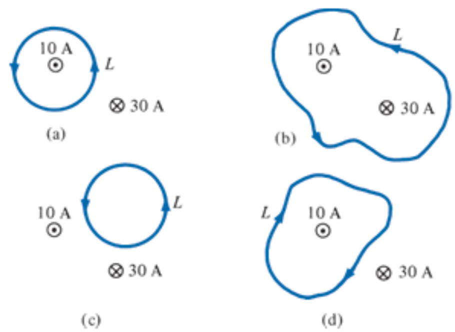


FIGURE 7.26 For Review Question 7.6.

7.8 Two identical coaxial circular coils carry the same current I but in opposite directions. The magnitude of the magnetic field **B** at a point on the axis midway between the coils is

(a) Zero
(b) The same as that produced by one coil
(c) Twice that produced by one coil
(d) Half that produced by one coil.

7.9 Which one of these equations is not Maxwell's equation for a static electromagnetic field in a linear homogeneous medium?

(a) $\nabla \cdot \mathbf{B} = 0$
(b) $\nabla \times \mathbf{D} = 0$
(c) $\oint_L \mathbf{B} \cdot d\mathbf{l} = \mu_o I$
(d) $\oint_S \mathbf{D} \cdot d\mathbf{S} = Q$
(e) $\nabla^2 \mathbf{A} = \mu_o \mathbf{J}$

7.10 Two bar magnets with their north poles having strength $Q_{m1} = 10 \text{ A} \cdot \text{m}$ and $Q_{m2} = 10 \text{ A} \cdot \text{m}$ (magnetic charges) are placed inside a volume as shown in Figure 7.27. The magnetic flux leaving the volume is

(a) 200 Wb
(b) 30 Wb
(c) 10 Wb
(d) 0 Wb
(e) −10 Wb

Answers: 7.1c, 7.2c, 7.3 (a)-(ii), (b)-(vi), (c)-(i), (d)-(v), (e)-(iii), 7.4d, 7.5a, 7.6 (a) 10 A, (b) −20 A, (c) 0, (d) −10 A, 7.7b, 7.8a, 7.9e, 7.10d.

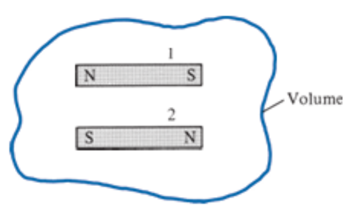


FIGURE 7.27 For Review Question 7.10.

Chapter 8

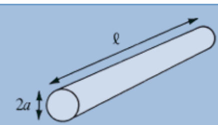
TABLE 8.1 Force on a Charged Particle

State of Particle	E Field	B Field	Combined E and B Fields
Stationary	$Q\mathbf{E}$	—	$Q\mathbf{E}$
Moving	$Q\mathbf{E}$	$Q\mathbf{u} \times \mathbf{B}$	$Q(\mathbf{E} + \mathbf{u} \times \mathbf{B})$

TABLE 8.3 A Collection of Formulas for Inductance of Common Elements

1. Wire

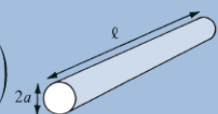
$$L = \frac{\mu_o \ell}{8\pi}$$



2. Hollow cylinder

$$L = \frac{\mu_o \ell}{2\pi}\left(\ln \frac{2\ell}{\mathbf{a}} - 1\right)$$

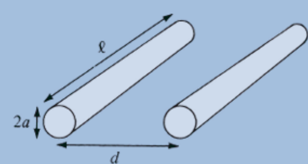
$$\ell \gg \mathbf{a}$$



3. Parallel wires

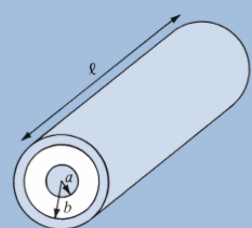
$$L = \frac{\mu_o \ell}{\pi} \ln \frac{d}{a}$$

$$\ell \gg d,\ d \gg a$$



4. Coaxial conductor

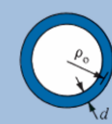
$$L = \frac{\mu_o \ell}{2\pi} \ln \frac{b}{a}$$



5. Circular loop

$$L = \frac{\mu_o \ell}{2\pi}\left(\ln \frac{4\ell}{d} - 2.45\right)$$

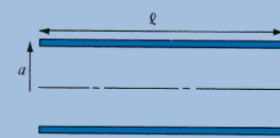
$$\ell = 2\pi\rho_o,\ \rho_o \gg d$$



6. Solenoid

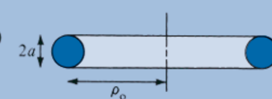
$$L = \frac{\mu_o N^2 S}{\ell}$$

$$\ell \gg a$$



7. Torus (of circular cross section)

$$L = \mu_o N^2[\rho_o - \sqrt{\rho_o^2 - a^2}]$$



8. Sheet

$$L = \mu_o\, 2\ell\left(\ln \frac{2\ell}{b + t} + 0.5\right)$$

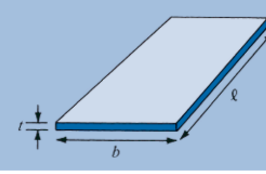


TABLE 8.4 Analogy between Electric and Magnetic Circuits

Electric	Magnetic
Conductivity σ	Permeability μ
Field intensity E	Field intensity H
Current $I = \int \mathbf{J} \cdot d\mathbf{S}$	Magnetic flux $\Psi = \int \mathbf{B} \cdot d\mathbf{S}$
Current density $J = \frac{I}{S} = \sigma E$	Flux density $B = \frac{\Psi}{S} = \mu H$
Electromotive force (emf) V	Magnetomotive force (mmf) $\mathcal{F}$
Resistance R	Reluctance $\mathcal{R}$
Conductance $G = \frac{1}{R}$	Permeance $\mathcal{P} = \frac{1}{\mathcal{R}}$
Ohm's law $R = \frac{V}{I} = \frac{\ell}{\sigma S}$	Ohm's law $\mathcal{R} = \frac{\mathcal{F}}{\Psi} = \frac{\ell}{\mu S}$
or $V = E\ell = IR$	or $\mathcal{F} = H\ell = \Psi\mathcal{R} = NI$
Kirchhoff's laws:	Kirchhoff's laws:
$\sum I = 0$	$\sum \Psi = 0$
$\sum V - \sum RI = 0$	$\sum \mathcal{F} - \sum \mathcal{R}\Psi = 0$

SUMMARY

1. The Lorentz force equation

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{u} \times \mathbf{B}) = m\frac{d\mathbf{u}}{dt}$$

relates the force acting on a particle with charge Q in the presence of EM fields. It expresses the fundamental law relating EM to mechanics.

2. Based on the Lorentz force law, the force experienced by a current element $I\,d\mathbf{l}$ in a magnetic field $\mathbf{B}$ is

$$d\mathbf{F} = I\,d\mathbf{l} \times \mathbf{B}$$

From this, the magnetic field $\mathbf{B}$ is defined as the force per unit current element.

3. The torque on a current loop with magnetic moment $\mathbf{m}$ in a uniform magnetic field $\mathbf{B}$ is

$$\mathbf{T} = \mathbf{m} \times \mathbf{B} = IS\mathbf{a}_n \times \mathbf{B}$$

4. A magnetic dipole is a bar magnet or a small filamental current loop; it is so called because its $\mathbf{B}$ field lines are similar to the $\mathbf{E}$ field lines of an electric dipole.

5. When a material is subjected to a magnetic field, it becomes magnetized. The magnetization $\mathbf{M}$ is the magnetic dipole moment per unit volume of the material. For linear material,

$$\mathbf{M} = \chi_m \mathbf{H}$$

where χ_m is the magnetic susceptibility of the material.

6. In terms of their magnetic properties, materials are either linear (diamagnetic or paramagnetic) or nonlinear (ferromagnetic). For linear materials,

$$\mathbf{B} = \mu\mathbf{H} = \mu_o\mu_r\mathbf{H} = \mu_o(1 + \chi_m)\mathbf{H} = \mu_o(\mathbf{H} + \mathbf{M})$$

where μ = permeability and $\mu_r = \mu/\mu_o$ = relative permeability of the material. For nonlinear material, $B = \mu(H)\,H$, that is, μ does not have a fixed value; the relationship between B and H is usually represented by a magnetization curve.

7. The boundary conditions that **H** or **B** must satisfy at the interface between two different media are

$$\mathbf{B}_{1n} = \mathbf{B}_{2n}$$

$$(\mathbf{H}_1 - \mathbf{H}_2) \times \mathbf{a}_{n12} = \mathbf{K} \;\; or \;\; \mathbf{H}_{1t} = \mathbf{H}_{2t}, \;\text{if}\; \mathbf{K} = \mathbf{0},$$

where $\mathbf{a}_{n12}$ is a unit vector directed from medium 1 to medium 2.

8. Energy in a magnetostatic field is given by

$$W_m = \frac{1}{2}\int_v \mathbf{B} \cdot \mathbf{H}\, dv$$

For an inductor carrying current I

$$W_m = \frac{1}{2} LI^2$$

Thus the inductance L can be found by using

$$L = \frac{\int_v \mathbf{B} \cdot \mathbf{H}\, dv}{I^2}$$

9. The inductance L of an inductor can also be determined from its basic definition: the ratio of the magnetic flux linkage to the current through the inductor, that is,

$$L = \frac{\lambda}{I} = \frac{N\Psi}{I}$$

Thus by assuming current I, we determine **B** and $\Psi = \int_s \mathbf{B} \cdot d\mathbf{S}$, and finally find $L = N\Psi/I$.

10. A magnetic circuit can be analyzed in the same way as an electric circuit. We simply keep in mind the similarity between

$$\mathcal{F} = NI = \oint \mathbf{H} \cdot d\mathbf{l} = \Psi\mathcal{R} \quad \text{and} \quad V = IR$$

that is,

$$\mathcal{F} \leftrightarrow V,\ \Psi \leftrightarrow I,\ \mathcal{R} \leftrightarrow R$$

Thus we can apply Ohm's and Kirchhoff's laws to magnetic circuits just as we apply them to electric circuits.

11. The magnetic pressure (or force per unit surface area) on a piece of magnetic material is

$$P = \frac{F}{S} = \frac{1}{2}BH = \frac{B^2}{2\mu_o}$$

where B is the magnetic field at the surface of the material.

12. Magnetic levitation (maglev) is a way of using EM fields to levitate objects. One important area of application of maglev is transportation. Conventional railroads operate at speeds below 300 km/hr, while maglev vehicles are designed for operating speeds of up to 500 km/hr.

Chapter 9

SUMMARY

1. In this chapter, we have introduced two fundamental concepts: electromotive force (emf), based on Faraday's experiments, and displacement current, which resulted from Maxwell's hypothesis. These concepts call for modifications in Maxwell's curl equations obtained for static EM fields to accommodate the time dependence of the fields.
2. Faraday's law states that the induced emf is given by $(N = 1)$

$$V_{\text{emf}} = -\frac{\partial \Psi}{\partial t}$$

For transformer emf, $V_{\text{emf}} = -\int \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$

and for motional emf, $V_{\text{emf}} = \int (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$.

3. The displacement current

$$I_d = \int \mathbf{J}_d \cdot d\mathbf{S}$$

where $\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$ (displacement current density) is a modification to Ampère's circuit law. This modification, attributed to Maxwell, predicted electromagnetic waves several years before the phenomenon was verified experimentally by Hertz.

4. In differential form, Maxwell's equations for dynamic fields are:

$$\nabla \cdot \mathbf{D} = \rho_v$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Each differential equation has its integral counterpart (see Tables 9.1 and 9.2) that can be derived from the differential form by using Stokes's theorem or the divergence theorem. Any EM field must satisfy the four Maxwell's equations simultaneously.

5. Time-varying electric scalar potential $V(x, y, z, t)$ and magnetic vector potential $\mathbf{A}(x, y, z, t)$ are shown to satisfy wave equations if Lorenz's condition is assumed.
6. Time-harmonic fields are those that vary sinusoidally with time. They are easily expressed in phasors, which are more convenient to work with. The cosine reference, can be used to show that the instantaneous vector quantity $\mathbf{A}(x, y, z, t)$ is related to its phasor form $\mathbf{A}_s(x, y, z)$ according to

$$\mathbf{A}(x, y, z, t) = \text{Re}[\mathbf{A}_s(x, y, z)\, e^{j\omega t}]$$

Chapter 10

SUMMARY

1. The wave equation is of the form

$$\frac{\partial^2 \Phi}{\partial t^2} - u^2 \frac{\partial^2 \Phi}{\partial z^2} = 0$$

with the solution

$$\Phi = A \sin(\omega t - \beta z)$$

where u = wave velocity, A = wave amplitude, ω = angular frequency $(=2\pi f)$, and β = phase constant. Also, $\beta = \omega/u = 2\pi/\lambda$ or $u = f\lambda = \lambda/T$, where λ = wavelength and T = period.

2. In a lossy, charge-free medium, the wave equation based on Maxwell's equations is of the form

$$\nabla^2 \mathbf{A}_s - \gamma^2 \mathbf{A}_s = 0$$

where $\mathbf{A}_s$ is either $\mathbf{E}_s$ or $\mathbf{H}_s$ and $\gamma = \alpha + j\beta$ is the propagation constant. If we assume $\mathbf{E}_s = E_{xs}(z)\,\mathbf{a}_x$, we obtain EM waves of the form

$$\mathbf{E}(z, t) = E_o e^{-\alpha z} \cos(\omega t - \beta z)\mathbf{a}_x$$

$$\mathbf{H}(z, t) = H_o e^{-\alpha z} \cos(\omega t - \beta z - \theta_\eta)\mathbf{a}_y$$

where α = attenuation constant, β = phase constant, $\eta = |\eta|\underline{/\theta_\eta}$ = intrinsic impedance of the medium, also called wave impedance. The reciprocal of α is the skin depth $(\delta = 1/\alpha)$. The relations between β, ω, and λ as stated in item 1 remain valid for EM waves.

3. Wave propagation in media of other types can be derived from that for lossy media as special cases. For free space, set $\sigma = 0$, $\varepsilon = \varepsilon_o$, $\mu = \mu_o$; for lossless dielectric media, set $\sigma = 0$, $\varepsilon = \varepsilon_o\varepsilon_r$, and $\mu = \mu_o\mu_r$; and for good conductors, set $\sigma \simeq \infty$, $\varepsilon = \varepsilon_o$, $\mu = \mu_o$, or $\sigma/\omega\varepsilon \to \infty$.

4. A medium is classified as a lossy dielectric, a lossless dielectric, or a good conductor depending on its loss tangent, given by

$$\tan\theta = \frac{|\mathbf{J}_s|}{|\mathbf{J}_{d_s}|} = \frac{\sigma}{\omega\varepsilon} = \frac{\varepsilon''}{\varepsilon'}$$

where $\varepsilon_c = \varepsilon' - j\varepsilon''$ is the complex permittivity of the medium. For lossless dielectrics $\tan\theta \ll 1$, for good conductors $\tan\theta \gg 1$, and for lossy dielectrics $\tan\theta$ is of the order of unity.

5. In a good conductor, the fields tend to concentrate within the initial distance δ from the conductor surface. This phenomenon is called the skin effect. For a conductor of width w and length ℓ, the effective or ac resistance is

$$R_{ac} = \frac{\ell}{\sigma w \delta}$$

where δ is the skin depth.

6. The Poynting vector $\mathcal{P}$ is the power-flow vector whose direction is the same as the direction of wave propagation; its magnitude is the same as the amount of power flowing through a unit area normal to the propagation direction.

$$\mathcal{P} = \mathbf{E} \times \mathbf{H}, \quad \mathcal{P}_{\text{ave}} = 1/2 \operatorname{Re}(\mathbf{E}_s \times \mathbf{H}_s^*)$$

7. If a plane wave is incident normally from medium 1 to medium 2, the reflection coefficient Γ and transmission coefficient τ are given by

$$\Gamma = \frac{E_{ro}}{E_{io}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}, \quad \tau = \frac{E_{to}}{E_{io}} = 1 + \Gamma$$

The standing wave ratio, s, is defined as

$$s = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

8. For oblique incidence from lossless medium 1 to lossless medium 2, we have the Fresnel coefficients as

$$\Gamma_{\|} = \frac{\eta_2 \cos\theta_t - \eta_1 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}, \quad \tau_{\|} = \frac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$$

for parallel polarization and

$$\Gamma_{\perp} = \frac{\eta_2 \cos\theta_i - \eta_1 \cos\theta_t}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t} \quad \tau_{\perp} = \frac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$$

for perpendicular polarization. As in optics,

$$\theta_r = \theta_i$$

$$\frac{\sin\theta_t}{\sin\theta_i} = \frac{\beta_1}{\beta_2} = \sqrt{\frac{\mu_1 \varepsilon_1}{\mu_2 \varepsilon_2}}$$

Total transmission or no reflection ($\Gamma = 0$) occurs when the angle of incidence θ_i is equal to the Brewster angle.

9. Microwaves are EM waves of very short wavelengths. They propagate along a straight line like light rays and can therefore be focused easily in one direction by antennas. They are used in radar, guidance, navigation, and heating.